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ON

Artificial Intelligence: The Gap Between Expectations and Real-World Performance

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In

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DECLARATION

We hereby certify that the work which is being presented in the project report entitled "Artificial Intelligence: The Gap Between Expectations and Real-World Performance" in partial fulfilment of requirement for the award of the degree of Bachelor of Engineering (Computer Science and Engineering) submitted in the department of Computer Science and Engineering at Chitkara University Institute of Engineering and Technology, Chitkara University, Punjab, India, is an authentic record of our own work carried out under the supervision of the designated mentor. The matter presented in this project report has not been submitted in any other university/institute for the award of any degree.

Place: Rajpura

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This is to certify that the above statement made by the candidates is correct to the best of my knowledge and belief.

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Finally, we extend our gratitude to our families and peers for their unwavering moral support during the course of this work.

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ABSTRACT

Artificial Intelligence (AI) has become a major topic of discussion in the past few years in the industry. This massive excitement brought in more than \$180 billion of investment in AI between 2024 and 2025 from private funding alone. These huge investments helped with many research studies during this time period that led to frontier models that achieved excellent scores across reasoning, coding, and multimodal benchmarks. Major organisations, companies and even government sectors started integrating AI into their work systems with expectations like fast-paced work, reduced labour, better productivity and large-scale automation. But reality had a different story to tell.

By 2026, many of these expectations were not fully achieved. It was not that AI was not improving, but it was more like an integration issue. Integrating these AI systems into existing workflows in these organisations became a much bigger challenge than anticipated. This difference between the expectations from AI and the reality we witnessed is called the AI Reality Gap — the gap which is the centre of all research and findings in this report.

This research report dwells on the main three reasons behind the gap. The first is the Human Verification Tax — the time and money spent by humans while checking the outputs of AI systems. The second is Enterprise Data Readiness — the unclean and unorganised data in organisations that leads to degraded AI performance. The third is Organisational Resistance — where companies struggle to change their workflows, integrate AI systems and retrain their employees.

A reference case study (Mayo Clinic Augmented Intelligence Programme) and an original Cross-Model Hallucination experiment were conducted to investigate these three factors. Technical implementations — a Data Quality Degradation experiment and a Productivity J-Curve mathematical model — were used to quantify the gap. The final conclusion is that the AI Reality Gap is not a model problem but a systems problem arising from data quality, human oversight, and the way AI systems are integrated into organisations.

1. Introduction

1.1 The Rise of AI Optimism (2024–2025)

There has been significant growth in the usage of AI in both public and private sectors between 2024 and early 2025. AI was seen as the next major technological revolution because of fast-paced improvement in Large Language Models (LLMs) and other AI systems. According to the Stanford AI Index (2025), global private investment in AI crossed \$180 billion and the United States was the largest investor. AI was described as a General-Purpose Technology (GPT) that could change many industries.

Many big organisations started to explore how AI could help with automation of many crucial tasks, decision-making and reducing operational overhead. AI was expected to become something more than just simple chatbots — something capable of handling difficult tasks and managing workflows with very little human intervention. AI systems started to show excellent results in reasoning, coding, and multimodal abilities, and because of this many people developed optimistic belief that AI could impact knowledge-based professions at a large scale in the near future.

1.2 The Emerging AI Reality Gap (2026)

However, by 2026, the optimistic belief of many around the globe about AI shifted to more careful and realistic discussions because AI did not match expectations when it came to real-world impact. AI systems were used by many companies in one or more business areas, but it was still a challenge to measure large-scale productivity gains. This situation is often called the productivity paradox — where the benefits of new technologies take more time than anticipated to show results.

AI systems have become significantly useful for tasks like writing reports, summarising documents and helping with programming. However, these improvements were not always reflected in overall organisational efficiency. Although many companies have successfully built prototypes, integration and implementation of AI systems for full production deployment remain challenges in complex business environments. This mismatch between what was expected and what we got is the AI Reality Gap. This mismatch matters a lot because in a lab, a 90% accuracy rate is an A+, but in a bank or a hospital (real world), a 5% to 10% error is a catastrophe.

1.3 Research Objectives

This report examines three factors that contribute to the AI Reality Gap:

- **The Human Verification Tax:** The time and cost associated with human supervision of AI outputs.
- **Enterprise Data Readiness:** The limitations of legacy data infrastructure in supporting AI models.
- **Organisational Resistance:** Challenges in redesigning workflows to accommodate AI integration.

2. Methodology

This section explains the research method used to study the difference between how artificial intelligence performs in test environments and how accurate it is in real situations. The study uses two case studies: a previously conducted real-world implementation of AI in healthcare, and a Cross-Model Hallucination case study conducted as part of this research.

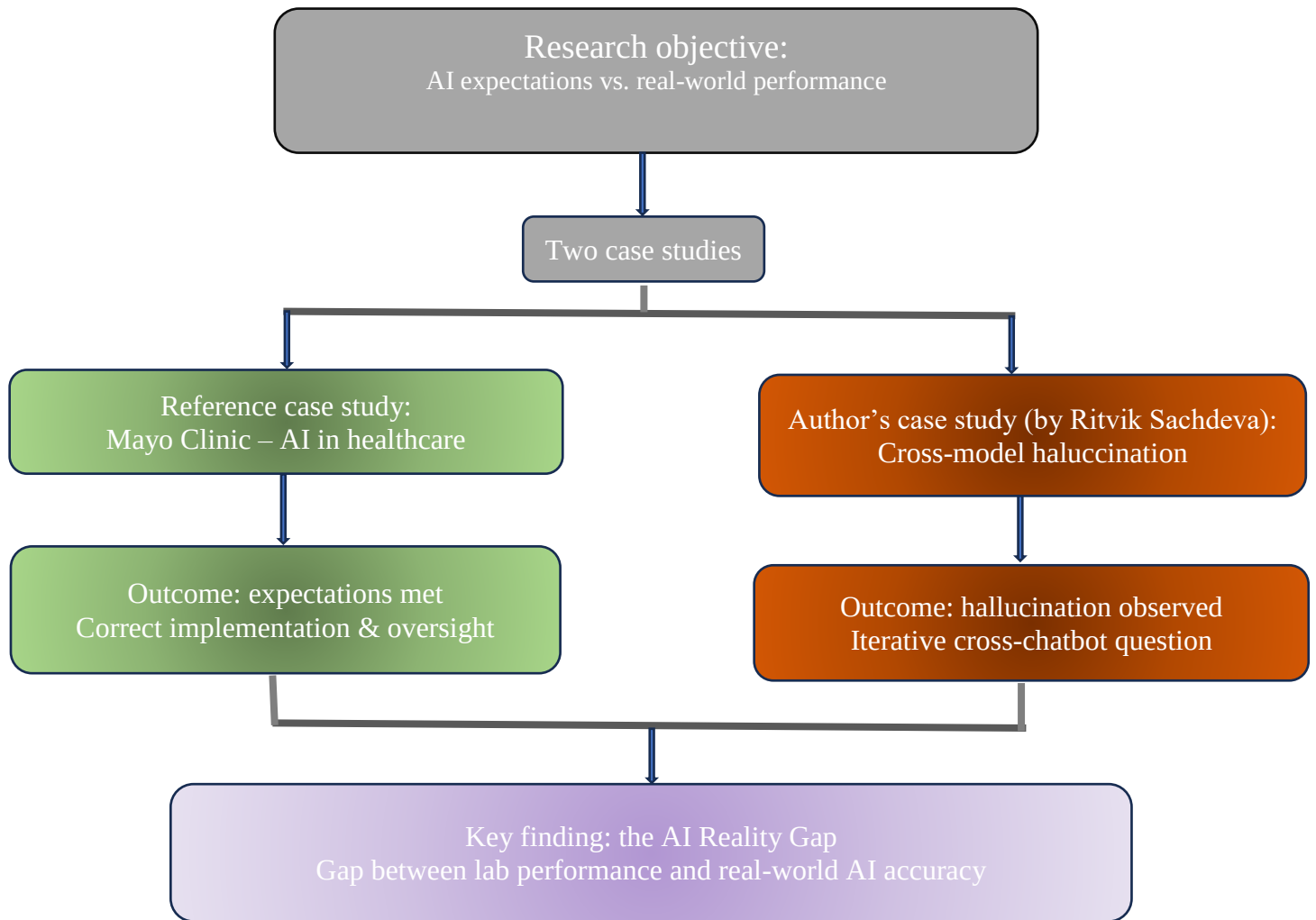


Figure 1 Research Methodology Overview — Two Case Studies

2.1 Reference Case Study: Augmented Intelligence in Healthcare (Mayo Clinic)

This section describes the implementation of Augmented Intelligence systems developed at the Mayo Clinic to help doctors with medical decision-making. This case study demonstrates how AI can successfully assist professionals when it is integrated carefully into existing workflows. The purpose of including this previously conducted case study is to show how correct implementation, proper data usage and human supervision can help AI systems perform more reliably and produce useful results in real-world environments.

In healthcare institutions, a large number of medical records — including medical history, diagnostic images, ECGs, and test results — are manually analysed on a daily basis by medical professionals, which is a very time-consuming process. Researchers at Mayo Clinic developed AI systems that can analyse medical data and assist doctors in real-time to identify hidden health issues and predict possible medical outcomes. These systems are referred to as Augmented Intelligence — AI designed to assist doctors, not replace them.

Important system design principles included:

- AI models trained using large datasets consisting of medical records and ECG data.
- Integration of these systems into the existing clinical workflow so that doctors could use them during diagnosis and treatment planning.
- Doctors remained responsible for reviewing and confirming the insights provided by the AI system, ensuring human supervision at every stage.

The Augmented Intelligence systems were mainly used for assistance with medical decision-making and for diagnostics like analysing electrocardiogram (ECG) data. Studies reported that these AI-assisted ECG analyses were able to detect certain heart abnormalities with around 93% accuracy, helping doctors detect hidden and potential health risks at a much earlier stage. These AI tools also assisted with clinical documentation and data management, reducing the workload of doctors and allowing them to spend more time on patient care.

2.2 Author Case Study: Cross-Model Hallucination Experiment

As part of this research, a controlled experiment was conducted by author Ritvik Sachdeva to study hallucination — a common issue faced while using AI systems. The experiment used two AI chatbots, ChatGPT and Google AI, and iterative questioning eventually led to a hallucination in one of the chatbots.

Initially, Google AI was asked to generate Python coding problems specially designed to test ChatGPT's logic-reasoning and problem-solving capabilities. The questions generated were progressively more complex, beginning with basic closure and string manipulation problems and escalating to a grid-robot simulation with wraparound and prime-coordinate twist rules. ChatGPT answered all questions correctly. However, when Google AI was informed of ChatGPT's Java translation of the final solution rather than being asked to solve the original problem directly, it produced a hallucinated response — confidently incorrect — as documented in the experiment's figures.

The Cross-Model Hallucination case study illustrates one of the practical limitations of AI systems: hallucination can occur not from data noise but from indirect reasoning conditions. This reflects the AI Reality Gap because a 90% lab accuracy does not prevent an AI from producing confident wrong answers in chained, real-world reasoning pipelines.

```

// Java
public class RobotSimulator {

    // Helper function to check if a number is prime
    public static boolean isPrime(int n) {
        if (n < 2) return false;
        for (int i = 2; i * i <= n; i++) {
            if (n % i == 0) return false;
        }
        return true;
    }

    public static String simulateRobot(String commands) {
        String[] directions = {"N", "E", "S", "W"}; // North, East, South, West
        int[] dx = {0, 1, 0, -1}; // change in x
        int[] dy = {1, 0, -1, 0}; // change in y

        int x = 0, y = 0;
        int dirIdx = 0; // start facing North

        for (int i = 0; i < commands.length(); i++) {
            char cmd = commands.charAt(i);

            if (cmd == 'L') {
                dirIdx = (dirIdx + 3) % 4; // turn left
            } else if (cmd == 'R') {
                dirIdx = (dirIdx + 1) % 4; // turn right
            } else if (cmd == 'M') {
                x = (x + dx[dirIdx] + 10) % 10; // move with wraparound
                y = (y + dy[dirIdx] + 10) % 10;

                // Prime twist: extra right turn if x+y is prime
                if (isPrime(x + y)) {
                    dirIdx = (dirIdx + 1) % 4;
                }
            }
        }

        return "(" + x + ", " + y + ", " + directions[dirIdx] + ")";
    }

    public static void main(String[] args) {
        String commands = "MMRMMMLMM";
        String finalPosition = simulateRobot(commands);
        System.out.println(finalPosition); // Output: (2, 9, S)
    }
}

```

Figure 2 Response from ChatGPT

Try this spatial-logic coding challenge:

"Write a Python function to simulate a robot on a 10x10 grid. The robot starts at (0,0) facing North. It follows these commands: 'L' (turn left 90°), 'R' (turn right 90°), and 'M' (move forward 1 step).

The grid has 'wraparound' (like Pac-Man): if it moves off the top, it appears at the bottom.

The Twist: Every time the robot lands on a coordinate where $x + y$ is a prime number, it must immediately execute an extra 'R' (turn right) before the next command.

Provide the final coordinates and facing direction for the command string: "MMRMMMLMM."

Why this usually fails:

1. **State Tracking:** It has to update (x, y) and direction simultaneously.
2. **The "Hidden" Trigger:** It often forgets to check the "Prime Number" rule *after every single move*, usually only checking it at the very end of the string.
3. **The Wraparound:** It might forget the `modulo 10` logic during the intermediate steps.

Figure 3 Question given by Google in AI mode

AI Mode ▾ All Videos Images Forums More ▾

This is where the hallucination hides! The Java code it wrote is logically sound, but its **execution simulation** is wrong. ChatGPT calculated the result as **(2, 9, S)**, but if you actually run that exact code, the answer is different.

Let's trace the robot's path manually (the "Prime Twist" changes everything):

1. **Start:** (0,0) Facing N.
2. **M:** (0,1). $0 + 1 = 1$ (Not Prime).
3. **M:** (0,2). $0 + 2 = 2$ (Prime!). Robot turns Right. Now facing East.
4. **R:** (Command says turn Right). Now facing South.
5. **M:** (0,9) [Wraparound]. $0 + 9 = 9$ (Not Prime).
6. **M:** (0,8). $0 + 8 = 8$ (Not Prime).
7. **M:** (0,7). $0 + 7 = 7$ (Prime!). Robot turns Right. Now facing West.
8. **L:** (Command says turn Left). Now facing South.
9. **M:** (0,6). $0 + 6 = 6$ (Not Prime).
10. **M:** (0,5). $0 + 5 = 5$ (Prime!). Robot turns Right. Now facing West.

Actual Result: (0, 5, W)
GPT's Claim: (2, 9, S)

Why it failed:
The AI is great at writing the *template* for the logic, but it cannot "run" the code in its head perfectly. It likely simulated the first few steps, got overwhelmed by the state changes, and then "guessed" a coordinate that looked plausible. It essentially hallucinated the **output** of its own perfectly valid code.

Figure 4 Hallucinated response from Google AI

public class RobotSimulator {

// Helper function to check if a number is prime

public static boolean isPrime(int n) {

if (n < 2) return false;

for (int i = 2; i * i <= n; i++) {

if (n % i == 0) return false;

}

return true;

}

public static String simulateRobot(String commands) {

String[] directions = {"N", "E", "S", "W"}; // North, East, South, West

int[] dx = {0, 1, 0, -1}; // change in x

int[] dy = {1, 0, -1, 0}; // change in y

int x = 0, y = 0;

int dirIdx = 0; // start facing North

for (int i = 0; i < commands.length(); i++) {

char cmd = commands.charAt(i);

if (cmd == 'L') {

dirIdx = (dirIdx + 3) % 4; // turn left

} else if (cmd == 'R') {

dirIdx = (dirIdx + 1) % 4; // turn right

} else if (cmd == 'M') {

x = (x + dx[dirIdx] + 10) % 10; // move with wraparound

y = (y + dy[dirIdx] + 10) % 10;

// Prime twist: extra right turn if x+y is prime

if (isPrime(x + y)) {

dirIdx = (dirIdx + 1) % 4;

}

}

return "(" + x + ", " + y + ", " + directions[dirIdx] + ")";

}

public static void main(String[] args) {

String commands = "MMRMMMLMM";

String finalPosition = simulateRobot(commands);

System.out.println(finalPosition); // Output: (2, 9, S)

}

STDIN

Input for the program (Optional)

Output:

(2, 9, S)

Figure 5 Code run in a compiler proving Google AI Hallucination and ChatGPT correct

3. Tools and Technologies

3.1 Machine Learning Framework

The primary machine learning implementation used in this research is the Random Forest classifier, an ensemble method comprising 100 decision trees. The implementation was carried out using the Scikit-learn library (Python), which provided robust and reproducible model training, evaluation, and noise injection capabilities.

3.2 Development Environment

All experiments were implemented in Python on Google Colab, a cloud-based Jupyter Notebook environment. Google Colab was chosen because it is freely accessible, reproducible, and supports the full scientific Python stack without local installation. All code is fully reproducible and can be re-executed independently.

3.3 Dataset

The UCI Heart Disease dataset from the Cleveland Clinic Foundation was used for the Data Quality Degradation experiment. The dataset contains 303 patient records with 13 key clinical features including age, chest pain type, blood pressure, cholesterol, ECG results, and maximum heart rate. It is publicly available through the UCI Machine Learning Repository. The dataset mirrors the type of structured clinical data used by the Mayo Clinic, making it appropriate for validating the experiment's relevance to real-world healthcare AI deployment.

3.4 Data Sources for J-Curve Modelling

The parameters for the Productivity J-Curve mathematical model were anchored to two primary data sources:

- Stanford AI Index 2025 — for AI adoption rates, investment figures, and benchmark progression data.
- McKinsey & Company — The State of AI in 2024 — for enterprise adoption statistics and AI ROI failure rates.

3.5 AI Systems Evaluated

Two large language model chatbot systems were used in the Cross-Model Hallucination experiment:

- ChatGPT (OpenAI) — served as the problem-solving model that received progressively complex Python questions.
- Google AI (Gemini) — served as the question-generating and evaluation model; the hallucination was observed in its responses.

4. Implementation

4.1 Experiment 1 — Data Quality Degradation

The first experiment quantifies the Enterprise Data Readiness factor by measuring the impact of data quality degradation on a Random Forest classifier trained on the UCI Heart Disease dataset.

Dataset Preprocessing

Three preprocessing steps were applied before training:

- Mean imputation: Missing values in the num_vessels and thal columns were filled using column means, illustrating the Human Verification Tax at a data-preparation level.
- Z-score normalisation: Features were scaled to zero mean and unit variance ($x' = (x - \mu)/\sigma$) to prevent high-range features such as cholesterol from biasing the model.
- Stratified train-test split: 80% of data was used for training and 20% for testing, with stratification on the target variable to preserve class balance.

Noise Injection

To mimic real-life enterprise data issues, a noise injection function was applied to both training and test data at intensity levels from 0% to 100% in 5% steps. Three noise types were used:

- Missing value noise: Sets a proportion p of random cells to zero, simulating incomplete records from manual data entry errors or broken data transfers.
- Corruption noise: Replaces a proportion p of cells with random values from $N(0, 64)$, introducing extreme outliers similar to wrong units, typing errors, or faulty sensors.
- Mixed noise: Applies both missing and corruption noise in equal parts — the most realistic condition, reflecting enterprise data that is simultaneously incomplete and corrupted.

A separate, freshly trained Random Forest model was used at each noise level to prevent information leakage between experiments.

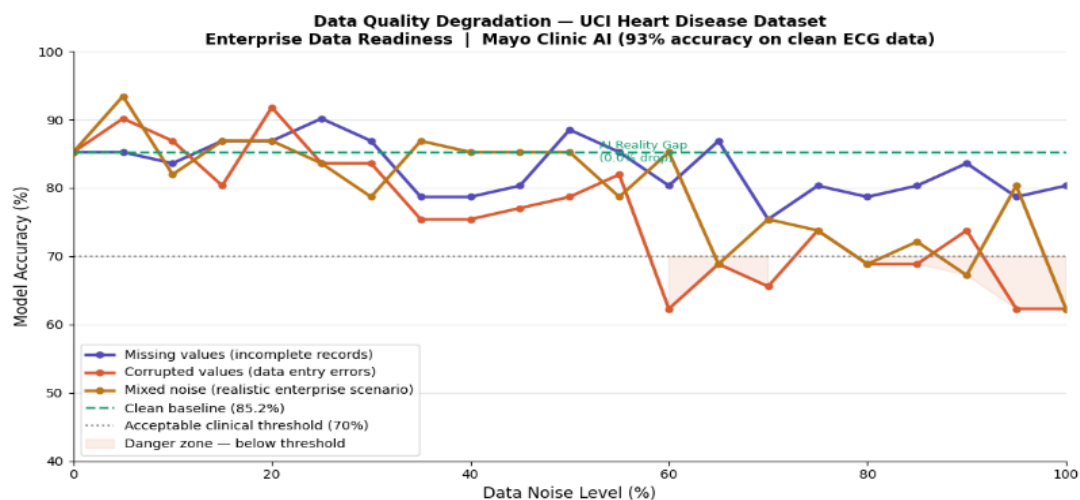


Figure 6 Data Quality Degradation

4.2 Experiment 2 — Productivity J-Curve Mathematical Model

The second experiment formalises the Productivity J-Curve as a continuous-time differential equation to model the productivity impact of AI adoption under three distinct organisational scenarios.

The J-Curve was expressed as:

$$P(t) = B - D \cdot e^{(-\delta t)} + G \cdot (1 - e^{(-\gamma t)})$$

Where the parameters are defined as follows:

Symbol	Meaning
B	Pre-AI baseline productivity, normalised to 100
D	Depth of the initial productivity drop
δ	How quickly the disruption fades
G	Maximum long-term productivity gain
γ	How quickly gains accumulate

Table 1: J-Curve Model Parameter Symbols and Meanings

Three scenarios were modelled:

- Scenario 1 — Ideal Adoption (Mayo Clinic): $D=6$, $G=32$, $\gamma=0.18$. Low disruption because AI is used as a support tool; high, fast gain matching the Mayo Clinic's reported outcomes.
- Scenario 2 — Typical Enterprise Adoption: $D=18$, $G=22$, $\gamma=0.08$. Moderate disruption from workflow redesign and retraining; slower gains reflecting Stanford AI Index findings on large organisations.
- Scenario 3 — Proof-of-Concept Trap: $D=28$, $G=8$, $\gamma=0.04$. High, slow-fading disruption due to over-ambitious scope; minimal long-term gain reflecting McKinsey data on failed AI ROI projects.

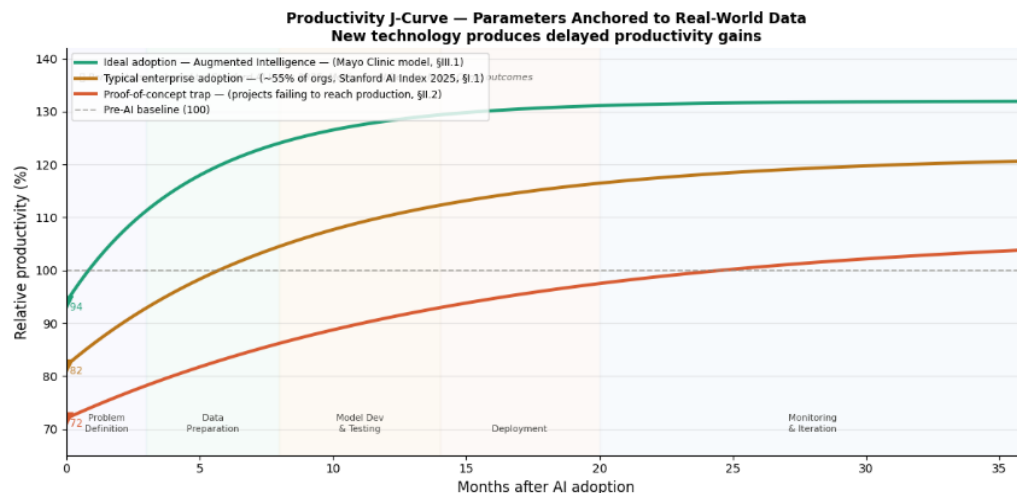


Figure 7 Productivity J-curve

5. Major Findings / Outcomes / Results

5.1 Data Quality Degradation Results

The Random Forest classifier trained on clean, pre-processed UCI Heart Disease data achieved a baseline accuracy of 85.25%. This is the model's lab benchmark — its performance when inputs are neat and controlled. The table below summarises accuracy at key noise intensities:

Noise Level	Missing Value Noise	Corruption Noise	Mixed Noise
0% (clean)	85.25%	85.25%	85.25%
25%	86.89%	83.61%	85.25%
50%	81.97%	78.69%	85.25%
75%	81.97%	68.85%	63.93%
90%	70.49%	60.66%	68.85%
100%	78.69%	65.57%	59.02%

Table 2: Accuracy vs. Noise Level across Three Noise Types

Missing Value Noise: Accuracy degraded slowly, staying between 85–88% through the first 35% of noise before dropping steadily to its lowest point of 70.49% at 90% noise — barely above the 70% clinical acceptability threshold.

Corruption Noise: Accuracy fell more sharply. At 60% noise it dropped to 65.57% — below the clinical cutoff. Extreme outlier values mislead the model more severely than missing values, because a missing value is at least identifiable as absent.

Mixed Noise: The most dangerously deceptive pattern. Accuracy at 50% noise remained at 85.25% — identical to the clean baseline — then suddenly crashed to 63.93% at 75% noise and continued falling to 59.02% at full corruption. This delayed collapse means AI systems may appear fine for a long time before suddenly failing.

The 70% clinical acceptability threshold was crossed at around 90% for missing noise, around 60% for corruption noise, and around 75% for mixed noise. Enterprise data quality assessments regularly report that between 20% and 80% of records in legacy systems have some form of data quality issue. The mixed noise result at 75% — a drop of nearly 22 percentage points from the clean baseline — is a realistic estimate of what a clinical AI system could be operating at in a hospital that has not invested in data governance.

This finding directly quantifies the Human Verification Tax: every output produced below 70% accuracy is one where a physician cannot act on the AI recommendation without independent verification. As noise increases, the proportion of outputs requiring additional review increases — and so does the cost, time, and cognitive load placed on the human in the loop.

5.2 Productivity J-Curve Results

The three adoption scenarios produced distinct productivity trajectories over a 36-month observation window.

- Scenario 1 — Ideal Adoption: Productivity started at 94.0 at adoption (a drop of only 6 points from the pre-AI baseline of 100), reflecting low disruption when AI is used as a support tool. Recovery was fast and sustained, reaching 132.0 by month 36 — a net gain of 32% above the pre-AI baseline, consistent with the Mayo Clinic's reported outcomes.
- Scenario 2 — Typical Enterprise Adoption: Productivity started at 82.0 — an 18-point drop from baseline — due to more significant workflow redesign and retraining. Recovery was slower, reaching 120.6 by month 36, representing a genuine 20.6% gain above baseline, though difficult to directly attribute to AI in standard organisational metrics.
- Scenario 3 — Proof-of-Concept Trap: Productivity dropped the most sharply to 72.0 at adoption — 28 points below baseline. Recovery was slow throughout the observation window, finishing at only 103.9 by month 36 — just 3.9% above the pre-AI baseline after three years. In realistic terms, a 3.9% productivity gain after three years of investment and disruption is indistinguishable from failure.

The contrast between Scenario 1 and Scenario 3 at month 36 is 28.1 points — 132.0 versus 103.9. This gap exists entirely because of the conditions around the AI system, not because of any difference in the model itself. The AI Reality Gap is a systems problem, not a model problem.

5.3 Cross-Model Hallucination Finding

The hallucination experiment documented an instance of AI failure under indirect reasoning conditions — categorically different from data quality degradation. ChatGPT solved all progressively complex problems correctly, including the grid-robot simulation with prime-coordinate twist rules. However, when Google AI was asked to evaluate ChatGPT's output rather than solve the original problem directly, it produced a confident, incorrect response.

This finding is significant because organisations are not deploying single models in isolation — they are building pipelines where one model's output feeds another. The hallucination experiment demonstrates that chained reasoning introduces failures that are categorically different from data quality failures. They do not appear in accuracy metrics measured on clean input and do not improve with better data governance. They require mandatory human checkpoints at each stage of any multi-model chain.

5.4 Case Study Synthesis

The Mayo Clinic's 93% diagnostic accuracy on structured ECG data corresponds to the near-zero noise region of the Experiment 1 degradation curve — where data is complete, clean, and domain-specific. The degradation experiment shows that this level of performance is attainable and sustainable, but only when the surrounding conditions are engineered to preserve it.

The Mayo Clinic case is not an argument that AI in healthcare always performs well. It is an argument that AI works when the whole system around it — data quality, integration design, and human oversight — is engineered to match the conditions under which the model was validated. This is a far rarer condition than it might appear, and most organisations have not yet met this requirement.

6. Conclusion and Future Scope

6.1 Conclusion

This report investigates a specific question: why has AI, despite achieving record performance on benchmark evaluations and attracting over \$180 billion in global private investment, failed to deliver the transformative productivity gains widely anticipated by 2026? The answer is that the gap between expectation and reality is not primarily about what AI is capable of doing — it is about the conditions under which it is deployed.

The data quality degradation experiment demonstrated that a robust Random Forest classifier achieved 85.25% accuracy on clean data but fell to 59.02% under full mixed-noise corruption — a drop of more than 26 percentage points on real medical data at noise levels within the range routinely reported by enterprise data quality assessments. The AI Reality Gap is not closed by deploying a better model. It is closed by deploying a better data environment.

The Productivity J-Curve model produced a 28.1-point productivity gap at month 36 between ideal adoption (132.0) and the proof-of-concept trap (103.9) — using the same underlying technology under different organisational conditions. This is the paper's central argument expressed as a number: the AI Reality Gap is a management and infrastructure problem, and it is one that organisations have the capacity to address today.

The cross-model hallucination experiment documented a dimension of the AI Reality Gap that data governance alone cannot address. Even with correct input and clean reasoning, AI systems can fail when the reasoning task is indirect or layered — precisely the conditions of real-world multi-model pipelines. Human oversight must be a principle, not an afterthought.

The tools and knowledge required to close the AI Reality Gap already exist. All that remains is the recognition by organisations working on AI adoption that this issue is their responsibility — not the model's.

6.2 Future Scope

Three research directions follow naturally from this work:

- **Retrieval-Augmented Generation (RAG) Experiment:** Comparing a grounded model against a plain LLM would measure how much of the AI Reality Gap grounding alone can close — directly addressing the hallucination finding.
- **Longitudinal Organisational Study:** Tracking AI adoption across enterprises over 24 to 36 months would empirically validate or challenge the J-Curve model's predictions with direct observational evidence rather than survey-based parameters.
- **Cross-Domain Noise Injection:** Applying the noise injection methodology to datasets from finance, legal, and manufacturing would test whether the degradation patterns found in clinical data generalise across industries or vary by domain.

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8. Appendices

Appendix A: Benchmark Progression Data (Stanford AI Index 2025)

The following benchmark improvements were recorded between 2023 and 2025 for frontier AI models:

- MMMU (Massive Multi-Discipline Multimodal Understanding and Reasoning Benchmark): +18.8%
- GPQA (Graduate-Level Google-Proof Q&A Benchmark): +48.9%
- SWE-bench (Software Engineering Benchmark): +67.3%

These gains in controlled benchmark settings stand in contrast to the mixed productivity outcomes observed in enterprise deployments, as quantified by the Productivity J-Curve model in Section 4.2 and discussed in Section 5.2.

Appendix B: Mayo Clinic Augmented Intelligence — Key Reported Metrics

The following figures were reported from the Mayo Clinic's Augmented Intelligence programme for ECG-based cardiac diagnosis, as referenced in §III.1:

- **Diagnostic accuracy:** 93% on structured ECG data with human oversight
- **Data type used:** 12-lead electrocardiogram (ECG) recordings processed daily
- **Integration model:** AI as decision-support layer within existing clinical workflow — physicians retained final decision authority at every stage
- **Human oversight:** Mandatory physician review of all AI-generated diagnostic flags before clinical action
- **Key outcome:** Earlier detection of cardiac abnormalities including atrial fibrillation, with measurable reduction in physician documentation workload
- **Data conditions:** Structured, consistently formatted, domain-specific clinical inputs — conditions approximating the clean baseline (0% noise) scenario in Experiment 1

These figures are drawn from Nair et al. (2022) as cited in Reference [4] of this paper.

Appendix C: Cross-Model Hallucination — Problem Sequence

The sequence of Python problems used in the Cross-Model Hallucination experiment, in order of increasing complexity:

- Closure problem: Create a list of 5 functions using a for loop, where each function prints its own index (0–4).
- String manipulation: Reverse 'PythonIsFun', conditionally remove characters based on string length parity, then repeatedly append until the string is exactly 37 characters long.
- Self-referential function: Write a function `count_self` that prints the exact number of characters in its own source code without reading a file.

- Grid-robot simulation: Simulate a robot on a 10×10 wraparound grid with an additional rule that the robot executes an extra right-turn whenever it lands on a coordinate whose x+y sum is a prime number. Compute final position for command string 'MMRMMMLMM'.

ChatGPT answered all four problems correctly. The hallucination in Google AI was triggered when it was asked to evaluate ChatGPT's Java translation of the grid-robot solution rather than solve the original problem directly.

Appendix D: Experiment 1 — Data Degradation Data

Baseline accuracy on clean data: **85.25%**.

- Under **missing value noise**, accuracy held stable through 35% noise before declining, reaching its lowest point of **70.49% at 90% noise** — just above the clinical threshold.
- Under **corruption noise**, the decline was steeper, crossing the 70% clinical threshold at approximately 60% noise and reaching a low of **60.66% at 90% noise**.
- Under **mixed noise**, accuracy held near baseline through 50% noise before collapsing sharply to **63.93% at 75%** and **59.02% at full corruption** — the most dangerous pattern of the three, as the delayed collapse makes early detection difficult.

Appendix E: Experiment 2 — J-Curve Scenario Parameters

The parameter values used for each adoption scenario in Experiment 2 are listed below:

- Scenario 1 (Ideal / Mayo Clinic): $B=100$, $D=6$, $\delta=0.20$, $G=32$, $\gamma=0.18$
- Scenario 2 (Typical Enterprise): $B=100$, $D=18$, $\delta=0.12$, $G=22$, $\gamma=0.08$
- Scenario 3 (Proof-of-Concept Trap): $B=100$, $D=28$, $\delta=0.07$, $G=8$, $\gamma=0.04$